UNIVERSAL AI UNIVERSITY

DEPARTMENT OF FUTURE TECH — RESEARCH PROPOSAL

Project Title:	Multi-Scale Overlapping Pattern Tokenization (MSOPT) for Non-Stationary Financial Time Series
Lead Researcher:	**Suryaansh Prithvijit Singh** (Universal AI University)
Faculty Guide:	**Prof. Shivaji Pawar**, Faculty Guide / Supervisor, Dept. of Future Tech
Date & Status:	August 2026 | Proposal Pending Formal Department Approval

■ CORE PROBLEM STATEMENT

EXISTING TIME SERIES DEEP LEARNING ARCHITECTURES SUFFER FROM A FUNDAMENTAL PARADIGM FAILURE WHEN APPLIED TO NON-STATIONARY FINANCIAL MARKETS:

1. THE 1D SERIAL POINTWISE BLINDSPOT: Serial models (LSTMs, vanilla Transformers) evaluate scalar price points *x_t* sequentially. This destroys local visual shape context (inflections, wedges, double bottoms) and causes quadratic computational explosion $O(T^2)$, preventing long historical context processing.

2. THE RIGID UNIFORM PATCHING BOTTLENECK: Modern patch transformers (PatchTST) force rigid uniform sequence slicing ($P=16$). In non-stationary markets, fixed boundaries clip pattern inflections arbitrarily, enforce single-scale rigidity, and lack translation invariance—rendering them fragile during market regime shifts.

3. THE QUANTITATIVE VISUAL GAP: Human quantitative traders analyze market dynamics visually as **multi-scale 2D spatial chart primitives** (micro-spikes, daily consolidations, macro regimes). Existing machine learning paradigms force 1D serial vectors or rigid 1D uniform grids, failing to extract localized multi-scale visual tokens.

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1. Executive Summary & Abstract

Financial price series (equities, indices, derivatives) present one of the most challenging domains for deep learning due to extreme non-stationarity, low signal-to-noise ratios, and regime-shifting dynamics. Current models force an unviable tradeoff between point-wise serial processing (*O(T^2)* computational complexity with zero local shape context) and fixed-patch uniform slicing (boundary clipping distortion with zero multi-scale adaptation).

This research proposes **Multi-Scale Overlapping Pattern Tokenization (MSOPT)**—a novel paradigm that reformulates financial time series modeling. Rather than treating market data as a 1D scalar sequence, MSOPT transforms time series into a **2D Scale-Time Spatial Tensor of multi-scale overlapping receptive fields**. By combining dilated receptive field extraction (*w \in [4, 32], d \in [1, 4]*) with 1D-SAX symbolic discretization (mean + trend slope quantization), MSOPT constructs an interpretable, translation-invariant vocabulary of market pattern tokens ("words").

This proposal outlines the theoretical motivation, literature gap, mathematical framework, validation protocol, and expected departmental contributions of the MSOPT research program at **Universal AI University**.

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2. 2025–2026 Literature Landscape & Research Gap Analysis

Model Family	Published / Venue	Core Mechanism	Critical Limitation in Financial Series
PatchTST	Nie et al. (ICLR 2023)	Uniform fixed patching ($P=16$, $S=8$) + ViT	Rigid patch length; boundary clipping; zero multi-scale adaptation.
TimesNet	Wu et al. (ICLR 2023)	1D \to 2D Tensor Reshaping via FFT periods	Assumes stationary periodicity; fails on non-periodic, regime-shifting financial markets.
BORF	Spinnato et al. (IEEE 2024)	Dilated Receptive Fields + 1D-SAX Discretization	Bag-of-Words loss of temporal sequence order; no end-to-end gradient learning.
TS-BPE	Götz et al. (May 2025/2026)	Byte Pair Encoding subseries merge	**Non-overlapping partition**; boundary clipping; not finance-tuned; black-box vocabulary.
DPR	Zhong et al. (May 2026)	Dynamic Pattern Recalibration soft-routing	Recalibration layer on top of tokens; **not a token discovery mechanism**.
PATK	AAAI-26 (March 2026)	Physics-aware HMM elastic tokenization	**Non-overlapping partition**; general-purpose; not human-legible.

The Critical Gap

No existing framework unifies multi-scale dilated receptive field extraction, dense translation-invariant overlapping tokenization ($s=1$), 1D-SAX human-legible codebooks, and 2D scale-time spatial tensor representations within a non-stationary financial modeling paradigm.

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3. Proposed Solution: Multi-Scale Overlapping Pattern Tokenization (MSOPT)

MSOPT introduces a 4-stage pipeline designed specifically to solve the non-stationary financial pattern problem:

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1. RAW FINANCIAL MULTIVARIATE SERIES
(Log Returns, Parkinson Volatility, Relative Volume)
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▼
2. MULTI-SCALE DILATED RECEPTIVE FIELD SAMPLING
(Window w in {4,8,16,32}, Dilation d in {1,2,4}, Stride s=1)
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▼
3. THRESHOLDED 1D-SAX SYMBOLIC DISCRETIZATION
(Segment Mean Quantization a_mu + Segment Slope Quantization a_beta)
■
▼
4. 2D SCALE-TIME SPATIAL TENSOR MAPPING
[Y-axis = Receptive Field Scale (w,d), X-axis = Time Index t]
■
▼
5. 2D SPATIAL CONVOLUTION & TRANSFORMER ENCODER
(Position + Scale + Volatility Multi-Dimensional Embeddings)
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▼
6. DIRECTIONAL THRESHOLD & REGIME CLASSIFICATION
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4. Task Formulation: High-Signal Classification Targets (Fork B Alignment)

To avoid the low signal-to-noise ratio trap of raw point return regression ($\hat{y}_{t+1} \in \mathbb{R}$, where MSE loss forces trivial "flat" predictions), MSOPT evaluates two robust classification tasks:

- **Directional Move Threshold Classification (y_{dir})**:

$$y_{dir, t} = \begin{cases} +1 & \text{if } R_{t:t+H} > +\delta \cdot \sigma_t \\ -1 & \text{if } R_{t:t+H} < -\delta \cdot \sigma_t \\ 0 & \text{otherwise (Neutral/Noise)} \end{cases}$$

where $\delta = 0.5$ volatility units over horizon $H \in \{1, 5, 20\}$ days.

- **Volatility Regime Shift Classification (y_{vol})**:

$$y_{vol, t} = \mathbb{I}\left(\sigma_{t:t+H} > 1.5 \cdot \bar{\sigma}_{30}\right)$$

Predicting upcoming market volatility expansions and regime breakdowns.

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5. Primary Research Questions & Testable Hypotheses

- **RQ1**: *Does dense overlapping receptive field tokenization (s=1) preserve predictive pattern signal better than single-scale fixed patching (PatchTST) and non-overlapping adaptive partitions (TS-BPE)?*

- **H1**: MSOPT will achieve statistically significant out-of-sample Sharpe ratio improvement over PatchTST and TS-BPE due to 100% translation invariance.

- **RQ2**: *Does 1D-SAX symbolic discretization improve signal-to-noise ratio in financial returns compared to continuous vector embeddings?*

- **H2**: Quantizing local subseries into discrete mean+slope symbols eliminates noise spikes while preserving structural shape primitives.

- **RQ3**: *Can a 2D Scale-Time Spatial Grid effectively model inter-scale pattern composition?*

- **H3**: 2D spatial convolutions across scale and time dimensions will isolate cross-scale market interactions (e.g., micro-spikes triggering macro-regime breaks) better than 1D temporal convolutions.

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6. Rigorous Methodology & Validation Protocol

To guarantee scientific validity and prevent backtest overfitting, MSOPT will be evaluated under a strict protocol:

- **Dataset**: 15+ years of high-liquidity market data (SP500 ETF `/SPY`, Apple `/AAPL`, Nasdaq ETF `/QQQ`, Treasury ETF `/TLT`) spanning multiple market regimes (2008 Financial Crisis, 2020 COVID shock, 2022 Inflationary bear market, 2023–2026 Tech bull market).
- **Walk-Forward Expanding Window Evaluation**: Initial 5-year training window, expanding annually. Zero lookahead bias.
- **Transaction-Cost Awareness**: All backtests enforce 5 bps slippage and commission costs per trade.
- **Benchmark Comparison**:

- *Baseline 1*: Standard Technical Features + LightGBM / XGBoost

- *Baseline 2*: PatchTST (Fixed Patch Transformer)

- *Baseline 3*: TimesNet (FFT 2D Transformer)

- *Baseline 4*: TS-BPE (Byte Pair Encoding Transformer)

- *Baseline 5*: Native BORF + Bag-of-Words Classifier

• **Evaluation Metrics**:

- Statistical: Directional Accuracy (%), AUC-ROC, Macro F1-Score.

- Financial: Out-of-Sample Sharpe Ratio, Sortino Ratio, Maximum Drawdown (Calmar Ratio).

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7. Project Timeline & Milestones

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■ PHASE 1: Literature Synthesis & Departmental Blueprint (COMPLETED) ■
■ - Deep-dive knowledge base across 8 core literature domains ■
■ - Departmental Research Proposal compilation ■
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■ PHASE 2: Formal Architectural Specification & Codebook Design (Weeks 1-2) ■
■ - Mathematical formalization of 1D-SAX codebook & 2D spatial tensor grid ■
■ - Baseline benchmark environment setup (Walk-forward backtest protocol) ■
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■ PHASE 3: Framework Construction & Controlled Experiments (Weeks 3-5) ■
■ - Implementation of MSOPT Tokenizer & 2D Scale-Time Spatial Embedder ■
■ - Controlled ablation studies (MSOPT vs PatchTST vs TS-BPE vs TimesNet vs BORF) ■
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■ PHASE 4: Empirical Validation, Paper Writing & Department Defense (Weeks 6-8) ■
■ - Cross-asset generalization experiments & transaction-cost backtesting ■
■ - Drafting academic manuscript for submission to peer-reviewed venue ■
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8. Expected Deliverables & Departmental Impact

• **Academic Paper**: A formal research paper targeted for top-tier conference/journal submission (*IEEE PAMI*, *ICLR*, *ACM KDD*, or *Journal of Financial Data Science*).

• **Open-Source Python Framework**: A robust, well-documented time series tokenization toolkit (`msopt-ts`) bridging traditional quantitative finance and modern deep learning.

• **Novel Intellectual Property**: A benchmarked, state-of-the-art multi-scale pattern recognition architecture tailored for high-noise non-stationary domains.

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9. Departmental Approval & Sign-Off Request

We respectfully request departmental review and approval to proceed with the formal research program outlined above.

Role	Name	Signature	Date
Lead Researcher / Author	**Suryaansh Prithvijit Singh** (Universal AI University)	______________________	____/____/2026
Faculty Guide / Supervisor	**Prof. Shivaji Pawar** (Universal AI University)	______________________	____/____/2026

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Faculty Guide / Supervisor	**Prof. Shivaji Pawar** Faculty Guide, Dept. of Future Tech Universal AI University	________________________	___ / ___ / 2026